RESEARCH ARTICLE

Certified Action Sets and Decision Limits in Surrogate-Assisted Engineering Optimization

Shu-Chuan Chu[a,*], Ya-Yu Zhang[a], Zhenyu Meng[b] and Jia Zhao[c]

[a]School of Computer Science, Nanjing University of Information Science and Technology, Nanjing 210044, Jiangsu, China; [b]Affiliation to be confirmed; [c]Affiliation to be confirmed

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ABSTRACT
A useful surrogate must support a credible evaluation decision, not only accurate predictions. We develop a conditional audit that distinguishes action changes, finite-horizon consequences and information costs. For a fixed-kernel Gaussian process with one unknown nonnegative response, enclosed affine posterior means define a response-domain outer set of possible next actions. An exact rational envelope computes the complete relaxed-band set in $O(M \log M)$ arithmetic operations, retains isolated ties and supplies at most two rival indices per exclusion. On 114 archived engineering branches, both declared response domains retain 2–7 of 1024 actions and include every stored next action. All 232,564 original exclusion proofs and binary coefficient systems are reverified. The envelope produces identical action sets with 3.19-fold faster set construction, or 2.62-fold including separate certificate replay, relative to the frozen pairwise implementation. These are computational-stage improvements. The engineering bounds remain too weak to certify any of 71 nonbaseline H1 contrasts; H5 adaptive allocation also gives no independent diagnostic advantage. Full deployment savings and physical accuracy of the decision margin remain unestablished. The study contributes an executable, proof-producing action-set operator and an engineering evaluation separating computational efficiency from decision sufficiency.

KEYWORDS
expensive engineering optimization; Gaussian processes; numerical certificates; affine envelopes; decision auditing

Preparation copy: author affiliations b and c remain to be supplied.

1. Introduction

Many engineering optimization problems depend on simulations, model training, or experiments whose evaluations dominate the available computational or physical budget. Surrogate-assisted evolutionary algorithms and Bayesian optimization reduce this burden by fitting an inexpensive predictive model and deciding which candidate should receive the next exact evaluation (Jin 2011; Liang et al. 2025; Jones, Schonlau, and Welch 1998; Jin et al. 2019; Li, Zhan, and Zhang 2021). Model management is therefore the operational core of surrogate assistance: surrogate prediction is converted into an acquisition or management decision, that decision allocates a paid evaluation, and the

CONTACT Shu-Chuan Chu. Email: 100130@nuist.edu.cn

resulting label changes the subsequent optimization state. A reliable predictor is useful only when this chain improves the use of the available budget.

This distinction matters directly in simulation-based engineering design. A monitoring signal can redirect finite-element analyses without improving structural performance, even when its aggregate error diagnostic appears satisfactory. Reliable regional Kriging search has connected uncertainty to restricted search regions and infill decisions (Suprayitno and Yu 2019), while diagnostic benchmarking has exposed differences between standard-test and real-application algorithm performance (Zatarain Salazar et al. 2025). These studies motivate inspection of the route from surrogate information to paid engineering evaluations. The present beam application examines this route within a constrained design family; broader relevance to materials, manufacturing, robotics, and neural architecture search remains motivation rather than application validation in this study.

Recent methods use Gaussian-process uncertainty, ensembles, local models, temporal information, or adaptive infill rules to balance exploitation and exploration (Zou et al. 2025; Ma et al. 2025; Liu et al. 2025; Zhou et al. 2025; Yu et al. 2026; Liu et al. 2021). Online and conformal calibration research asks whether predictive distributions or bounds retain their intended reliability under sequential selection (Deshpande, Marx, and Kuleshov 2024; Kim et al. 2025; Jin and Ren 2024; Nardone et al. 2026). Calibration alone, however, does not establish that a signal changes candidate ordering, selects a better candidate at the same state, or improves a closed-loop trajectory after the surrogate and candidate distribution adapt. These properties form a decision pathway and must be evaluated separately.

The main computational object is a certified response-domain outer set of possible next actions for a fixed-kernel Gaussian process. Numerical coefficient bands account for the binary linear system; exact envelope queries produce closed feasible intervals and short exclusion witnesses. The engineering evaluation places this operator inside a conditional first-action audit: a learned state and objective context are fixed, the first action is varied, and each branch follows a declared continuation with the same external candidate stream and its own observed history. The horizon, comparison margin and reader-request budget specify the judgment. The audit differs from changing an entire policy throughout an optimization trajectory.

The evaluation separates computational correctness and cost from decision sufficiency. For the operator, we check complete action-set equality, closed ties and every exclusion witness, then compare construction times on identical coefficient bands. For downstream engineering use, we ask whether narrower action information changes a certifiable judgment and whether complete cost improves on the strongest simple same-target reader. Calibration, conditional intervention and whole-policy validation remain separate evidence objects. The implemented audit record, source-specific settings and frozen observation schedules are specified in Table 3.

The outputs are an executable conditional record, a same-target comparison with class-aware fixed depth, and explicit request/physical-attempt/execution ledgers. The independent primary record contains no better h5 alternatives and demonstrates no adaptive-allocation advantage. Exact caching, structural aliases and the immediate-query accounting proposition are implementation checks, rather than standalone optimization innovations. A descriptive opportunity analysis retains each dataset's original margin and every original seed group. Its supporting sources and the separate failed development stages appear in the supplement.

The study combines a proof-producing action-set operator with a bounded method evaluation. It does not establish a new acquisition rule, a positive-case diagnostic advantage, a superior selector or performance-preserving savings over complete common

continuation. The residual controller, immediate GP interventions and independent policy transfer provide complementary evidence within their stated targets; their reporting units are not pooled as independent samples. The computational contribution is an exact coefficient-band envelope with constant-size exclusion witnesses and a measured 3.19-fold construction improvement over the frozen pairwise reference. Its engineering evaluation supplies the boundary between sharper action information and a useful downstream judgment. The original learning-curve and audit failures remain part of the contribution's scope.

Reliability here means the credibility and utility of a surrogate-management decision, rather than structural failure-probability design.

2. Related Work

2.1. *Surrogate-assisted optimization and engineering applications*

Surrogate-assisted evolutionary computation has developed a broad range of models and management policies for expensive optimization (Jin 2011; Liang et al. 2025). Early frameworks combined evolutionary search with approximate fitness functions or Gaussian-process surrogates (Jones, Schonlau, and Welch 1998; Ong, Nair, and Keane 2003). Later work introduced selective ensembles, local/global models, classification-assisted search, and adaptive infill rules (Jin et al. 2019; Li, Zhan, and Zhang 2021; Wang, Jin, and Doherty 2017; Yu, Li, and Liang 2020; Sun et al. 2017). Differential evolution remains a common search engine in this area; the taxonomy and empirical evidence synthesized by Pan et al. (Pan et al. 2026a) show that exponential-crossover behavior depends strongly on operator design and parameter control. A recent survey identifies model construction and model management as separate determinants of performance (Liang et al. 2025). Controlled experiments with adjustable surrogate accuracy likewise show that prediction accuracy affects preselection, individual-based, and generation-based management differently (Hanawa, Harada, and Miura 2025).

Engineering applications show how these ideas enter expensive computational workflows. Hierarchical surrogate evaluation reduces neural-architecture search cost (Liu, Bao, and Leng 2025), deep surrogate models support multi-objective robotic linkage design (Ibrayev et al. 2024), and physics-informed Gaussian-process regression with adaptive Bayesian optimization combines finite-element knowledge with welding-parameter selection (Wang et al. 2025). Surrogate-assisted PSO has also been applied to multi-UAV path planning (Wang et al. 2026). These studies motivate reliability checks at the decision layer because prediction accuracy and evaluation savings are valuable only when surrogate outputs lead to useful paid evaluations.

Uncertainty-aware SAEAs typically make uncertainty candidate dependent. BGP-SAEA combines a boosting framework with Gaussian-process uncertainty (Zou et al. 2025), while recent large-scale methods use lower-confidence-bound Gaussian-process models and diversity search (Ma et al. 2025). For expensive multi-objective optimization, MLPSGP-SAEA combines multilayer-perceptron variable grouping, sparse Gaussian processes, and an adaptive sparse-and-diverse infill criterion (Pan et al. 2026b). Other work adapts differential-evolution selection to errors in both decision and objective spaces (Liu et al. 2025), tightens the search space for high-dimensional expensive problems (Zhou et al. 2025), or draws on multiple spatial, temporal, and performance information sources (Yu et al. 2026). These mechanisms can change candidate ranking, local search regions, or population updates.

Competitive SAEAs also use evolutionary populations and learned relations rather than a scalar acquisition alone. DRSO retains unevaluated promising solutions in a relation-model-assisted evolutionary loop to preserve useful population information under expensive-evaluation budgets (Hao, Zhang, and Zhou 2024). Its public author implementation makes it suitable for a source-frozen engineering comparison, while its distinct population mechanism broadens the evidence beyond Gaussian-process acquisitions.

2.2. *Reliability-aware surrogate model management*

Bayesian optimization has incorporated online calibration. Deshpande et al. (Deshpande, Marx, and Kuleshov 2024) calibrate predictive distributions and update them online. Localized online conformal Bayesian optimization constructs spatially varying thresholds and a calibrated posterior (Kim et al. 2025). Selection-conditional methods such as JOMI use reference sets under explicit invariance and exchangeability conditions (Jin and Ren 2024). Temporal conformal prediction has also been considered in surrogate-assisted evolutionary optimization (Nardone et al. 2026). The cross-model residual journal examined here does not establish the assumptions required by those methods and is treated as an empirical monitor without transferred guarantees.

Table 1. Scope of representative reliability and model-management approaches.

Approach	Primary quantity assessed	Direct decision question	Closed-loop consequence required
Gaussian-process uncertainty	Predictive confidence	Does uncertainty modify an acquisition value?	Usually evaluated by optimizer performance
Conformal prediction	Marginal or localized coverage	Does a calibrated set or bound satisfy its stated guarantee?	Not implied by coverage alone
Adaptive infill	Evaluation selection	Which candidate receives the next expensive label?	Evaluated through search performance
Candidate-dependent SAEA management	Ranking, region, or population update	Does surrogate information change the search action?	Method-specific
Relation-model DRSO	Pairwise fitness/category relations and unevaluated population	Does evolutionary surrogate management preserve promising search information?	Evaluated by paid-budget endpoint
Decision-consequence audit	Calibration, decision influence, utility, and final outcome	Does the signal change a paid decision and make it more useful?	Required as the final audit level

Several studies compare Bayesian, evolutionary, and hybrid surrogate strategies under constrained budgets (Gobert et al. 2024), and others show that unevaluated but surrogate-predicted candidates can influence evolutionary diversity (Hao, Zhang, and Zhou 2025). These optimizers are complementary to the present audit. A change in a coupled trajectory does not, by itself, isolate what the mechanism would have selected at the same learned state. The additional GP intervention freezes that state and candidate pool before consulting alternative outcomes, then evaluates longer-horizon policies separately.

2.3. *Operational distinction from existing model management and selection*

Reliable regional Kriging supplies a variance-guided search region and adaptive infill mechanism (Suprayitno and Yu 2019); surrogate-management frameworks can also provide convergence guarantees under their assumptions (Booker et al. 1999). The present method does not replace these search rules or inherit their convergence guarantees. It evaluates the route from an existing rule to a paid action and the evidence needed to justify that route. Hanawa et al. (Hanawa, Harada, and Miura 2025) vary the accuracy of a pseudo-surrogate to compare management strategies. Here the intervention instead holds an actually learned GP state fixed and varies its acquisition rule; posterior uncertainty and training labels are not replaced by synthetic accuracy levels.

Diagnostic benchmarking (Zatarain Salazar et al. 2025) studies effectiveness, efficiency, reliability, parameter controllability, and runtime dynamics on real applications. Our unit of intervention is the mechanism decision within a frozen optimization state, followed by a distinct policy-level validation. Algorithm selection and its data-acquisition cost are established topics (Kus et al. 2024; Kostovska et al. 2022). In particular, trajectory-based warm-starting reuses search information when switching algorithms (Kostovska et al. 2022). Our twenty-step selection workflow deliberately discards pilot labels from validation training to measure selection transfer under an intact information boundary. It is a diagnostic case, not a claimed advance over warm-starting selectors; a production workflow could reuse pilot information and would require its own validation.

Table 1a. Relation to representative prior work and the evidence object added here. The comparison concerns the stated studies, not the absence of these ideas throughout the literature.

Representative work	Reported focus	Operational evidence in the present implementation
Reliable regional Kriging (Suprayitno and Yu 2019)	Variance-guided reliable regions and sequential infill	Exact chosen-action equivalence and separate intervention labels at a frozen state
Surrogate management (Booker et al. 1999)	Managing approximation models with convergence analysis	An evidence audit without inheriting an optimizer's convergence guarantee
Adjustable-accuracy management study (Hanawa, Harada, and Miura 2025)	Search response to controlled pseudo-surrogate accuracy	Fixed learned posterior, identical candidate pool, and acquisition-only intervention
Diagnostic benchmarking (Zatarain Salazar et al. 2025)	Effectiveness, efficiency, reliability, parameter controllability, and runtime dynamics across real applications	Mechanism-level decision effects connected to separate closed-loop outcomes
Two-step lookahead (Wu and Frazier 2019)	Model-based expected future improvement for choosing evaluations	Recorded first actions followed by actual responses under a declared common continuation
Frugal selection (Kus et al. 2024) and trajectory warm-starting (Kostovska et al. 2022)	Selection data cost and reuse of search information	Independent pilot-to-validation transfer, exact paid ledger, and explicit endpoint/cost boundaries

The reusable technical object is the frozen decision record plus its equivalence classes, response ledger, and subsequent validation record. Neither logging, matched initialization, additive ranking invariance, nor counting diagnostic overhead is claimed as a standalone new idea. The combined record makes the signal-to-action claim and information boundary inspectable; the present results do not establish a diagnostic-quality increment over matched-state complete continuation. The evidence tests that method on a residual

controller and four GP acquisition policies; it does not establish general superiority across all surrogate or selector families.

3. Decision-Consequence Auditing Framework for Surrogate-Assisted Model Management

3.1. *Conditional first-action object*

The record is $(s, c, a, r, H, \tau, B, \mathcal{P}, \mathcal{E}, q, \mathcal{L})$: frozen learned state, objective context, first action, continuation, additional-action horizon, margin, request allowance, observed prefix, exact action class, output reason and cost ledger. With baseline mean first action and positive initial incumbent, $d_j(H) = [m_0(H) - m_j(H)]/m_{\text{init}}$. The independent primary target retains $H = 5$, $\tau = 0.001$ and $B = 18$ per state. The archived common continuation retains $\tau = 10^{-6}$. A complete branch has six response positions, h0 through h5. Equal endpoints do not define action equivalence. An immediate tie cannot certify a later tie, and an incomplete nonalias target abstains.

3.2. *Optimization system and information boundary*

A deterministic objective f(x) is minimized over a bounded continuous domain under budget B. At evaluation t, the labeled history contains only observations acquired before t. A surrogate fitted to that history supplies predictions for a candidate set, while a monitoring signal and model-management rule determine which candidate, region, evaluation gate, restart, or stopping action is used. The audit treats the full chain from surrogate output to paid evaluation as its object of analysis.

Before the objective is called, a prediction-before-label journal stores the candidate set or its reproducible identifier, selected point, surrogate outputs, monitoring signal, incumbent, decision mode, support indicators, and random-state identifiers. The record is flushed before the label is obtained. Labels acquired solely for frozen-state diagnostics remain separate and never return to the optimizer. This boundary prevents outcome information from leaking into the decision being audited.

The role of the framework in a surrogate-assisted optimization loop is shown in Figure 1.

Figure 1 locates the three audit levels along the operational chain from an expensive engineering problem to surrogate prediction, model-management choice, paid evaluation, and closed-loop outcome. Calibration assesses the signal encountered by the optimizer; decision analysis tests whether the signal changes a choice and improves common-state utility; closed-loop validation measures the final consequence after histories diverge.

3.3. *Certified response-domain action sets*

Consider n observed responses $y_1, \ldots, y_n$, a fixed first-action location with unknown response $z \geq \ell \geq 0$, and a finite eligible next-action pool of size M. Kernel parameters and the normalized noise diagonal are fixed. Let A be the augmented $(n+1)$-point binary kernel matrix and $w_i = A^{-1}k_i$. With target normalization, define $h_i = (1 - \mathbf{1}^{\mathsf{T}} w_i)/(n+1)$. The exact mean for this binary system is

$$\mu_i(z) = a_i z + b_i, \quad a_i = w_{i,n+1} + h_i, \quad b_i = \sum_{t=1}^{n} (w_{i,t} + h_i) y_t. \tag{1}$$

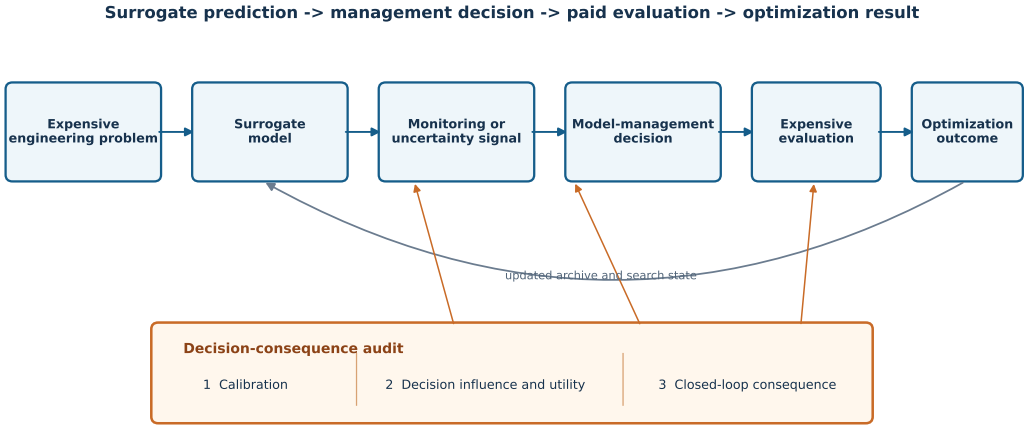


Figure 1. Decision-consequence auditing framework for surrogate-assisted optimization.

The target standard deviation cancels in the denormalized mean; this statement does not extend the operator to uncertainty-based acquisition rules. Linear posterior updates and affine-envelope geometry are established, including in knowledge-gradient computation (Frazier, Powell, and Dayanik 2009). The present operator concerns numerical coefficient enclosures, a closed response domain and verifiable exclusions, rather than a new acquisition policy.

The inverse-residual check bounds $\|I - A\widehat{P}\|_\infty \le \rho < 1$, giving $\|A^{-1}\|_\infty \le \|\widehat{P}\|_\infty/(1-\rho)$. A weight residual then bounds its coefficient error; normalization, dot products and endpoint construction are enclosed outwards. The stored implementation uses a stricter $\rho < 0.5$ gate. Failure retains the full eligible pool. This certificate is conditional on the supplied binary matrix and documented finite, normal arithmetic, with kernel evaluation error and native floating predictions at every unknown response outside its guarantee.

Write coefficient bands as $a_i \in [a_i^-, a_i^+]$, $b_i \in [b_i^-, b_i^+]$. For $z \ge 0$, let $L_i(z) = a_i^- z + b_i^-$ and $U_i(z) = a_i^+ z + b_i^+$. A potentially minimal action must satisfy

$$R_i = \{z \ge \ell : L_i(z) \le U_j(z) \text{ for every } j\} \ne \varnothing. \tag{2}$$

This is an exact relaxed-band problem and an outer approximation for the coupled GP coefficients. A retained action need not be realizable by those coupled coefficients.

Theorem 1 (closed-domain band filter). Given finite ordered binary bands and $\ell \ge 0$, all nonempty R_i, their exact rational endpoints and exclusions can be constructed in $O(M \log M)$ arithmetic operations and $O(M)$ storage. Every excluded action has an infeasibility witness using at most two rival indices, together with the domain constraint. Every minimizer allowed by the coefficient enclosures is retained, including actions feasible only at one response.

Proof. An allowed minimizer has $L_i \le \mu_i \le \mu_j \le U_j$ for every competitor, proving inclusion. Sort upper lines by decreasing slope and build their lower envelope $E(z) = \min_j U_j(z)$ with exact rational intersections. Parallel dominated lines can be removed from the envelope; a tied boundary value remains represented by its neighbors. Each line is inserted and removed at most once. For every original candidate, $E - L_i$ is a continuous concave piecewise-affine function with decreasing segment slopes. Binary search locates its maximum and the two possible zero crossings on $[\ell, \infty)$. A nonnegative maximum yields the closed feasible interval, including a singleton. A negative maximum

lies at the domain boundary, on a parallel segment or between two adjacent envelope lines. Those one or two constraints already make the candidate infeasible and supply its witness. Sorting costs $O(M \log M)$, the envelope scan $O(M)$, and the searches $O(\log M)$ per candidate. Rational bit complexity is additional. □

Algorithm 2. Exact band-envelope audit. (1) Enclose coefficients using the verified inverse gate. (2) Build the shared lower envelope of upper lines. (3) Query every original lower line, retaining all closed-domain ties. (4) Emit exact feasible intervals or at most two rival indices. (5) Independently replay each exclusion as rational one-dimensional infeasibility. (6) Apply identical already available point bounds and account separately for GP construction, pruning, proof replay and physical work. The algorithm accepts no pending response. The synthetic unit checks compare complete intervals with an independent all-pairs rational oracle and include a boundary-only lowest-index minimizer.

For one continuation, the available pending-point bound p, next-point bound λ_i and feasible response interval R_i imply the joint incumbent lower bound

$$J = \min\left(m_{\text{init}}, \min_{i:R_i \neq \varnothing} \min\{\max(p, \inf R_i), \lambda_i\}\right). \tag{3}$$

The nonnegative-range reader additionally receives the same pending-point trial bound as the full-pool reader. A paid baseline endpoint and $[J, m_{\text{init}}]$ can certify a category only when the complete effect interval lies in its declared category. This H1 calculation does not certify a five-step adaptive history.

3.4. *Decision pathways and ranking invariance*

A monitoring mechanism can affect optimization through candidate-specific ranking, an evaluation gate, a restart or stopping rule, a change of search region or population, or a stochastic exploration controller. The first audit step identifies at least one such path before additional expensive experiments are authorized.

Lemma 1 (ranking invariance under a candidate-independent additive correction). At evaluation t, suppose two rules receive an identical indexed candidate set, use deterministic and identical tie-breaking, and are evaluated in exact arithmetic. If every supported candidate shares the same additive correction c_t, then minimizing $H_t(x) = \mu_t(x) + c_t$ returns the same minimizer index as minimizing $\mu_t(x)$.

Proof. For any indexed candidates x_i and x_j, $H_t(x_i) - H_t(x_j) = \mu_t(x_i) - \mu_t(x_j)$. Every strict comparison and exact tie is therefore preserved under the stated tie-breaking rule. The same indexed minimizer follows. □

Corollary 1. If two configurations also share initialization, candidate generation, exploration decisions, all random streams and stochastic operators, numerical precision, and optimizer-state updates, and differ only by replacing the common-correction minimizer with the mean minimizer, their trajectories are identical by induction over evaluations. Finite-precision agreement is an implementation property and is checked separately in the experiment audit.

This result identifies a structural limitation of candidate-independent corrections; it does not invalidate uncertainty-aware surrogate management in general. Candidate-dependent posterior uncertainty, expected improvement, lower-confidence bounds, support-dependent rules, evaluation gates, restarts, and stopping policies may create decision paths that Lemma 1 does not remove.

3.5. *Three audit levels and promotion gate*

Common-history calibration fits the same frozen model and candidate pool, then evaluates monitoring losses and failures without updating the search. Common-state decision analysis compares the action with and without the mechanism at the same state; when actions differ, separate diagnostic labels measure true one-step utility. Independent closed-loop experiments then allow each policy to update its own history and evaluate final optimization behavior. One-step improvement diagnoses immediate exploitation value but is not a necessary condition for useful exploration. A lower immediate gain therefore identifies a need for delayed-value assessment rather than, by itself, grounds for stopping an exploration mechanism. The following procedure clarifies the interpretation of audit stages; it does not retrospectively alter any frozen experimental protocol or statistical comparison.

Evidence stages for a surrogate-assisted model-management audit.

Step 1 — Specify. Freeze the surrogate, candidate generator, monitoring signal, management or acquisition rule, expensive objective context, optimizer budget, audit-request budget, target horizon and margin, and deterministic replay information. In the independent conditional implementation, h0 is the first action and H=5 denotes five further common-rule actions. Output: protocol and information boundary.

Step 2 — Trace. Follow the signal through ranking, evaluation, region, population, restart, stopping, or stochastic-exploration decisions. Gate: at least one candidate-dependent or otherwise actionable path.

Step 3 — Record and calibrate. Persist every decision before its label. If the mechanism makes a calibrated-error claim, assess that claim on appropriately separated diagnostic data. Gate: an intact information boundary and evidence supporting the stated claim. Mechanisms that make no calibration claim are not required to pass a coverage test.

Step 4 — Test common-state utility. Freeze optimization states and candidate pools, compare baseline and mechanism actions, and label only the prespecified diagnostic alternatives. Report decision-change frequency and immediate utility. For exploration, region, restart, or stopping mechanisms, prespecify a delayed-value assessment where needed. The implemented point-action intervention uses the same frozen training history, fitted kernel and first candidate pool, then the same declared continuation and candidate-seed stream with each branch updating its own history. For positive initial incumbent m_{init}, the recorded conditional target is $d_j(H) = [m_0(H) - m_j(H)]/m_{\text{init}}$, where m_0 is the baseline-branch incumbent. The primary implementation uses $H = 5$ and margin $\tau = 0.001$. Gate: an identifiable decision consequence and sufficient evidence to justify the next diagnostic expenditure; lower one-step improvement alone does not reject exploration.

Step 5 — Validate the closed loop. Run independent optimization on held-out instances or engineering conditions under matched optimizer budgets, including when exploration requires a delayed-value test. Record diagnostic calls separately and include them in any deployment-cost comparison. Output: final objective consequence and total cost boundary.

Step 6 — Decide. Promote, revise, stop, or retain an inconclusive decision using prespecified outcome tolerances and cost limits. Report the first unsupported claim and the evidence needed to resolve it. Promotion requires closed-loop evidence for the intended engineering use. Output: stage-wise audit record. The independent replay reports an exact branch alias, a measured target-horizon outcome, an h0 proxy, or an abstention because the target remains incomplete. For nonalias branches, better/tie/worse certification

requires both baseline and alternative to reach h5; an immediate tie does not certify a later tie. Table 2 maps this implementation to its recorded inputs and outputs.

Table 2. Implementation contract of the independent conditional audit. Values and information boundaries are taken from the frozen collection/replay code. This is a factual specification, not a new convergence result.

Object	Recorded implementation	Evidence boundary
Frozen state	Own 100-label history, fitted kernel, root candidate pool and first-action map	No alternative or probe response enters the frozen root fit
Continuation	Same declared mean or EI rule and candidate-seed stream; each branch updates its own history	The first action is the intervention; this target differs from whole-policy runs
Horizon	h0: first action; h5: five further common-rule actions	Six prefix requests per completed branch in this implementation
Target / margin	Baseline-minus-alternative final incumbent divided by initial incumbent; primary margin 0.001	Better/tie/worse refers to the declared conditional h5 target
Branch reuse	Same history, fitted kernel, action, continuation, context and seed stream	A same-point response may be shared more broadly; an equal utility does not merge branches
Budget	Every branch-step request counts, including cache reads; primary B=18	Not the number of additional physical attempts
Evidence output	Exact alias; measured h5 outcome; h0 proxy; budget-incomplete abstention	No partial trajectory certificate or general prediction guarantee

Algorithm 1. Frozen conditional first-action audit. Table 3 gives its exact observation schedules and outputs.

Table 3. Frozen v1 conditional audit: executable operation specification. This table describes the existing code, including the unsuccessful adaptive heuristic; it does not propose a new optimizer. OpenAI Codex (GPT-6) assisted the code-to-method transcription.

Operation	Exact implementation
State / target	Freeze the 100-label history, fitted ARD kernel, first candidate pool, context and first actions. Baseline is the mean first action. Each branch updates its own history under the same mean or EI continuation and candidate-seed stream. $h0$ is the intervention; $H = 5$ is five further actions. The primary target is $d_j = [m_0(5) - m_j(5)]/m_{\text{init}}$, margin $\tau = 0.001$.
Alias / access	Group exactly identical first actions only within the same state and declared continuation. If $K = 1$, output structural tie for all names with zero requests. Otherwise `observe(branch,step)` may reveal only the next prefix response. Each request costs one, including an exact cache read.
Fixed depth	Complete the baseline to step 5; then complete alternatives in original class order while budget remains. Original class insertion follows mean, EI, PI, LCB. Incomplete comparisons abstain. Full mode uses $6K$ requests as its allowance.
Breadth / proxy	Breadth loops over steps 0–5 and advances baseline then alternatives at each step. Immediate mode reads h0 for each class; its labels are proxies for the h5 target, not certificates.
Adaptive reads	Read h0 and h1 for all classes, baseline first; then complete the baseline. Rank alternatives by descending $p_j = \lvert m_0(1) - m_j(1)\rvert/m_{\text{init}} + \sigma_j^{\text{root}}/m_{\text{init}}$. Equal priorities use original class order. Advance each alternative to stages 3 and 5 before moving to the next. This priority is an observed-effect/uncertainty heuristic, not a confidence bound.
Measured outputs	A nonalias better/tie/worse certificate requires both baseline and alternative to reach h5. Use better if $d_j > \tau$, worse if $d_j < -\tau$, otherwise tie. Budget-incomplete output is abstain; an immediate tie never certifies a later tie.

Operation	Exact implementation
Acquisitions	Independent budget collection uses 1024 candidates, normalized targets, $\alpha = 10^{-6}$ and $\kappa = 2$. EI/PI use an improvement margin 0.05 times training-label standard deviation. ARD parameters are fitted once at the shared 50-label warm state (max 30 iterations), then fixed. These settings differ from the archived intervention and transfer studies.
Calibration proxy	Eight frozen-root probes give NRMSE (scaled by training-label standard deviation) and empirical coverage of $\mu \pm 1.6448536269514722\sigma$. The constructed gate requires NRMSE ≤ 0.25 and coverage ≥ 0.8. If eligible, classify the root-EI difference divided by m_{init} using the same margin; otherwise abstain. It predicts h5 utility without a horizon guarantee. The original comparator charges eight probes in all 30 states.
State selection	Start with mean and effect threshold τ. Visit EI, PI, LCB; replace the selected control only for label better and a strictly larger recorded effect. Thus exact ties retain the earlier control. Select only from the disclosed outputs, not oracle h5 truth.
Loss / averaging	State-selection loss is $100[m_{\text{selected}}(5) - \min_j m_j(5)]/m_{\text{init}}$. Average over all 30 states, including alias-only states. Complete truth is used only after the reader finishes, to evaluate the decision; it cannot change the observation trace. Ten shared seed blocks, not these dependent states or contrasts, are the independent units.

The framework can return a positive result. Classical GP expected improvement is a candidate-dependent positive control: both its predicted improvement and uncertainty vary across candidates, so its acquisition can change ranking and passes the structural decision-influence check. Passing that first check does not guarantee superior endpoints; it authorizes empirical utility and closed-loop evaluation.

3.6. *Diagnostic cost and scope of deployment claims*

The optimization budget and the audit budget are different quantities. The total physical-call cost includes new optimizer evaluations, additional diagnostic evaluations, and recovery or verification calls; previously paid deterministic labels reused from an archive are identified separately. A comparison of optimizer endpoints at equal budgets does not establish that the audit itself saves evaluations. In the independent replay, each branch-step request is charged, including a cache read; a completed branch needs $H + 1 = 6$ requests. Its primary budget 18 is per state. For K exact branch classes, full-baseline/full-alternative reporting can complete at most $\min(K - 1, \max(0, \lfloor B/(H + 1) \rfloor - 1))$ nonbaseline comparisons under this request contract. This elementary capacity does not bound additional physical solves or audits with a justified early certificate.

The independent internal validation uses 907,200 optimization evaluations and 1,990,656 additional diagnostic labels. Those full-pool diagnostics support mechanism analysis on inexpensive benchmark functions, but are not proposed as the default screening cost for expensive engineering simulations. A bounded engineering audit should prespecify states, label the union of selected alternatives where possible, and stop or report inconclusive when its diagnostic limit is reached. Any claim of net savings must compare the audited workflow with a prespecified unaudited workflow, including diagnostic overhead. Section 6.6 reports a complete paid-workflow comparison with endpoint losses; it does not establish performance-preserving net savings.

3.7. *Decision equivalence and the minimum immediate diagnostic budget*

For a frozen state s, let $a_j(s)$ be the exact point selected by policy j and let c identify the deterministic objective context: simulator, physical condition, parameterization, and numerical configuration. Policies are immediately decision-equivalent when they select the

same bitwise point under the same c. This is a local relation. Equivalent first actions do not make later policies equivalent, and labels must not be reused across changed physical conditions or noisy objectives without an additional justification. Whole conditional branches share a record only when the frozen history, fitted kernel, first action, objective context, continuation and deterministic seed stream also coincide; across other states, exact point/context equality permits response reuse only. The reported implementations evaluate point-acquisition rules; conditional restart, region and stopping interventions are not independently validated here.

Let X be the union of distinct context-point pairs selected by all prespecified states and policies, and let A contain the exact responses that the auditor is permitted to reuse after decisions have been frozen. For incumbent m_s and fixed positive scale v_s, the reported immediate utility is $u_j(s) = \max(0, m_s - f_c(a_j(s)))/v_s$. The additional deterministic query requirement is

$$N_{\mathrm{diag}} = |X \setminus A|.$$

Proposition 1 (minimum worst-case immediate diagnostic queries). Suppose the objective is an unrestricted deterministic oracle, all selected points and incumbents are frozen, and each unqueried response can independently lie on either side of at least one associated incumbent while all other responses remain fixed. Querying each member of X outside A once suffices to report every immediate utility. In the worst case, fewer queries cannot determine all utilities exactly.

Proof. Sufficiency follows by assigning the same exact response to every context-point equivalence class and reusing only permitted responses. For necessity, consider an unqueried member x in a class with incumbent m. Two objectives can agree on all available and queried responses while taking values $m - \delta$ and $m + \delta$ at x for $\delta > 0$ ($\delta < m$ for a positive objective); the resulting utilities differ. No procedure using only those shared observations can distinguish them. For a finite set of distinct points, disjoint smooth local interpolants can realize these assignments, so smoothness alone does not remove the ambiguity. Querying each undetermined member resolves it. A certified response bound or a restricted physical objective class may reduce this worst-case requirement; no such additional exact inference is used here. □

The proposition is an allocation rule for a specified exact diagnostic report, not a lower bound on all possible optimization audits. It does not prove that the queries improve search. Section 6.5 illustrates both the within-state equivalence reduction and the additional reduction from permitted source-response and diagnostic-cache reuse. The source label at a frozen original action is consulted only after the intervention choices are fixed; it never enters the predecision GP fit. All thresholds for a deployment decision must still be fixed for the intended engineering use before its validation outcomes are observed. The added transfer study supplies descriptive evidence, not a retrospectively chosen promotion threshold.

4. Experimental Design

4.1. *Representative monitored optimizer and search framework*

A temporal residual monitor is used as the representative audited mechanism. For a selected point x_t, the standardized signed residual is r_t=[y_t-mu_t(x_t)]/s_t, where

s_t is the standard deviation of the current training labels and is replaced by one if it is zero. Historical residuals retain their prediction-time models and scales. For candidate x, the monitor forms H_t(x)=mu_t(x)+s_t q_t(x), where q_t(x) is the nonnegative weighted 0.9 quantile of at most the latest 100 residuals. Temporal weights have a half-life of 25 evaluations; spatial variants use a Gaussian kernel and an effective-sample-size support guard of 20.

The temporal prototype PEM-T converts recent failures, normalized bound width, and evaluations without incumbent improvement into an exploration probability between 0.1 and 0.8. Exploration selects the candidate with the largest minimum distance to the archive. Otherwise, the smallest supported upper value is used when an improvement declaration is available and the monitor is not alarmed; the smallest predicted mean is used in the remaining cases. Fixed01 retains the same model and candidate process with exploration probability 0.1. Under the temporal bound, q_t and s_t are common to supported candidates, so Lemma 1 predicts that the upper-value and mean exploitation rankings coincide.

The internal optimizer uses a randomized Latin hypercube with 2D+20 initialization points. The first D+10 points train the initial model; the remaining D+10 are predicted before labeling and create startup residuals. Inputs are scaled to the unit cube. A regularized Gaussian radial-basis-function surrogate uses distance divided by the square root of D, length scale 0.25, ridge 10^-6, and standardized labels. At most 256 observations are retained through an elite-plus-farthest-point rule.

Each iteration proposes 256 candidates: 128 reflected differential-evolution proposals and 128 scrambled Sobol points. Duplicates and candidates within 10^-10 of evaluated points are replaced. Purpose-specific random streams depend on seed and evaluation index rather than the total budget.

4.2. *Development and independent validation*

Development uses eight prespecified BBOB functions, dimensions 10 and 20, instances 1-3, and three repeats. Thirteen internal configurations separate exploration, residual weighting, and feedback components. The budget is 10D. The required gate compares PEM-T with a fixed exploration control selected only within the development protocol. Failure of the gate prevents any optimization-superiority claim.

After the gate fails, the unchanged mechanism is frozen on all 24 noiseless BBOB functions, dimensions 10, 20, and 40, instances 11-13, and three repeats. Six internal configurations produce 3,888 runs and 907,200 paid evaluations. Three history policies produce 7,776 frozen states and 1,990,656 diagnostic labels. Development and validation estimates are never pooled.

The endpoint is the base-10 logarithm of final error relative to common initialization error, with numerical flooring at 10^-12. Repeats are summarized within function-instance cells, instances are averaged, and functions receive equal weight. Nine prespecified validation contrasts form one Holm family. Paired hierarchical bootstrap intervals resample functions, instances, and common repeat indices. These intervals summarize suite heterogeneity and are not presented as a population theorem for all black-box tasks.

4.3. *External and application evidence*

Sobol, CMA-ES, classical GP expected improvement, and author-code TuRBO-1 provide budget-matched descriptive references (Eriksson et al. 2019). They retain their own initialization and search mechanisms, so these comparisons are not same-history causal effects. In the finite-element application, GP expected improvement (GP-EI), probability of improvement (GP-PI), and the lower-confidence-bound form of GP-UCB are additionally run from the same initial observations as the internal methods. All three acquisitions use the same Matérn-5/2 Gaussian process, candidate pool, and local-search budget; GP-PI uses a margin equal to 0.01 times the observed response standard deviation, and GP-UCB uses a fixed confidence multiplier of 2.0. DRSO uses the authors' public relation-model-assisted evolutionary implementation at commit 03c19d4 (Hao, Zhang, and Zhou 2024). The population size is set to 58 solely to accept the common initial archive; its relation models, unevaluated-population update, reproduction, training cap, and remaining defaults are unchanged. Three LassoBench tasks provide application transfer under fixed data and splits (Sehic et al. 2022). The BBOB implementation and reporting follow the COCO comparison framework (Hansen et al. 2021). Complete tables, the 300-dimensional extension, resource measurements, sensitivity studies, interruption accounting, and file-level audits appear in the Supplementary Materials.

5. Mechanism Results

5.1. *Development gate and implementation invariants*

All 1,872 development optimization runs and 432 common-history runs complete without exceptions. PEM-T and the configuration that removes its upper-value mode have identical trajectories in all 144 matched pairs. Temporal monitoring with fixed 0.3 management likewise matches fixed 0.3 mixing in all 144 pairs. These are implementation confirmations of Lemma 1 and Corollary 1, rather than independent effective components.

The development-selected control is fixed 01. PEM-T is worse on all eight development functions in both dimensions. Its function-weighted log-relative endpoint difference is +0.2831 in D=10 and +0.1810 in D=20, where positive values favor fixed 01. The required optimization-benefit gate is therefore unmet before the independent validation begins.

5.2. *Independent endpoint and immediate decision effects*

Table 4 reports the nine prespecified validation comparisons. PEM-T increases final relative error in all three dimensions. The base-10 differences correspond to geometric error ratios of 1.468, 1.258, and 1.159 for dimensions 10, 20, and 40. The effect decreases with dimension but remains in the unfavorable direction for 60 of 72 function-level comparisons. Common-state feedback decisions also have lower true immediate improvement than mean-greedy decisions in every dimension.

Table 4. Prespecified independent validation contrasts. Endpoint values above zero favor fixed 01; immediate values below zero favor mean-greedy selection.

Dimension	Contrast	Estimate	95% bootstrap interval	Direction count	Holm p
10	PEM-T minus fixed01 endpoint	0.1668	[0.0749, 0.2391]	22 positive, 2 negative	5.36e-6

Dimension	Contrast	Estimate	95% bootstrap interval	Direction count	Holm p
20	PEM-T minus fixed01 endpoint	0.0998	[0.0467, 0.1905]	19 positive, 5 negative	0.00170
40	PEM-T minus fixed01 endpoint	0.0639	[0.0130, 0.1123]	19 positive, 5 negative	0.00170
10	Feedback minus mean immediate improvement	-0.0170	[-0.0431, -0.0067]	1 positive, 18 negative	0.00078
20	Feedback minus mean immediate improvement	-0.0158	[-0.0352, -0.0079]	0 positive, 21 negative	0.00048
40	Feedback minus mean immediate improvement	-0.0344	[-0.0630, -0.0100]	1 positive, 20 negative	0.00048

The function-level distribution underlying the aggregate endpoint contrasts is shown in Figure 2.

Figure 2 displays all 24 function-level effects separately for each dimension rather than only the suite average. Most bars lie above the zero line: 22 of 24 in dimension 10 and 19 of 24 in both dimensions 20 and 40. The few negative bars show that the result is not pointwise uniform, while the shrinking bar heights across panels agree with the smaller aggregate effect at higher dimension. The conclusion is therefore based on a broad directional pattern with disclosed exceptions, not on one dominant benchmark function.

5.3. *Calibration does not identify optimization utility*

PEM-T explores 39.7%, 36.4%, and 34.4% of post-initialization evaluations in dimensions 10, 20, and 40, compared with approximately 10% for fixed 01. Its observed selected-point upper-bound failure fractions are 0.1008, 0.0942, and 0.0964, close to the nominal monitoring level. Nevertheless, its endpoints and common-state decisions are worse. A near-target aggregate failure fraction therefore does not imply efficient budget allocation.

Improvement declarations provide a second example. PEM-T makes 31, 34, and 155 declarations across the three dimensions; 20, 27, and 30 are false. The corresponding pooled false-declaration fractions are 0.645, 0.794, and 0.194. A one-sided failure rate near 0.1 cannot control the conditional error among the small subset declared to improve

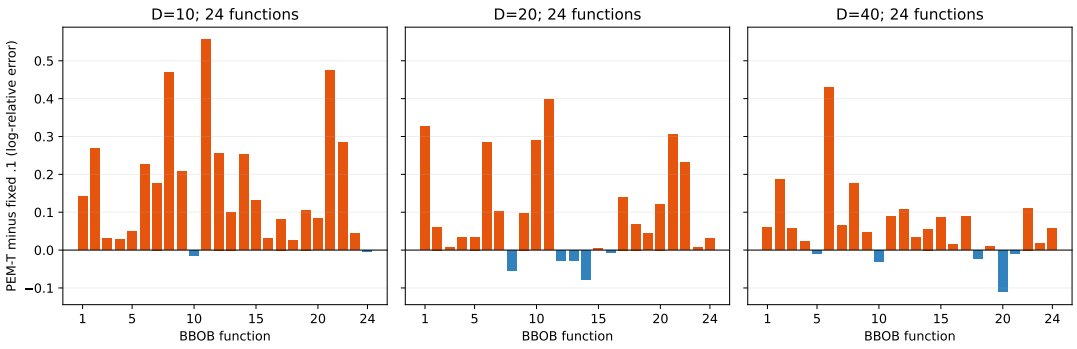


Figure 2. Function-level endpoint effects in the independent BBOB validation. Positive values indicate larger PEM-T errors.

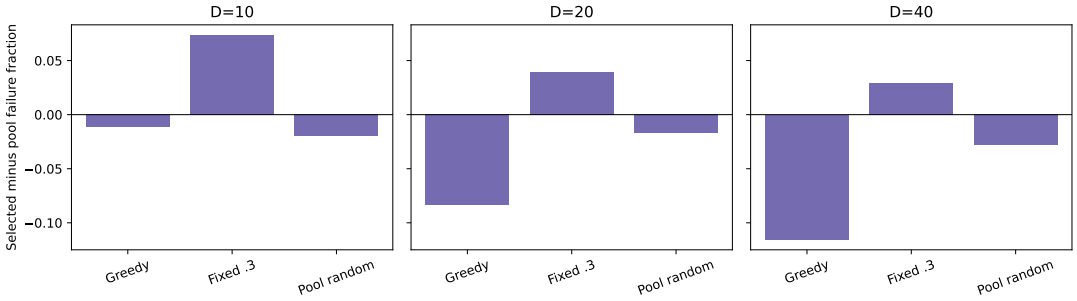


Figure 3. Selected-point minus pool failure differences under frozen history policies.

the incumbent.

History also changes the object being monitored. Under fixed 0.3 mixing, selected-point failures exceed pool failures, whereas under greedy histories the relationship reverses. The prespecified interaction is positive in all three dimensions and passes Holm adjustment. Selection-induced differences should therefore be attributed to the coupled history and policy, not treated as a universal direction of undercoverage.

The reversal of selected-point and pool failure differences across frozen history policies is shown in Figure 3.

Figure 3 makes the history dependence visible. Fixed 0.3 histories place the selected-point failure fraction above the pool value in all three dimensions, whereas greedy histories place it below the pool value. Random-pool histories remain closer to zero and are negative in each panel. Because the sign changes with the policy that generated the frozen state, a single universal "selection penalty' ' would misrepresent the monitored process.

Spatial-temporal weighting lowers selected-point quantile loss in the prespecified diagnostic comparisons, yet its closed-loop endpoint remains worse than fixed 01 in each dimension. This is the central empirical separation: a better residual estimate can still direct evaluations toward candidates with low optimization value.

To isolate the evidential role of each audit stage, a stage-omission ablation is constructed from the frozen records. Unlike an optimizer-component ablation, omitting an audit stage does not alter the evaluated trajectory; it removes the corresponding basis for inference. Table 5 states the strongest conclusion still supported at each stopping point and the evidence that would be lost.

Table 5. Stage-omission ablation and the resulting loss of evidence.

Available audit evidence	Apparent conclusion for the representative monitor	Information lost if the audit stops here
Calibration only	Selected-point failure is close to 0.1 and may appear satisfactory.	No evidence that exploitation ranking changes or that selected actions are useful.
Calibration plus ordering analysis	The temporal additive correction is exploitation-redundant; feedback can act through exploration.	The utility of the changed exploration decisions remains unknown.
Ordering plus common-state utility	Feedback changes actions, but those actions have lower immediate improvement than mean-greedy choices.	Delayed effects after histories diverge remain untested.

Available audit evidence	Apparent conclusion for the representative monitor	Information lost if the audit stops here
Full audit including closed loop	The controller does not pass the held-out optimization-benefit gate under the tested budgets.	No omitted audit stage is used to claim deployment value.

This evidence ablation uses the same frozen results rather than rerunning a modified optimizer because the ablated objects are inferential stages rather than search operators. It shows why a calibration-only assessment would have produced an incomplete reliability conclusion, why algebraic pathway analysis can identify redundancy without new objective calls, and why common-state and closed-loop evidence answer distinct questions.

6. Engineering Application: Surrogate-Assisted Structural Optimization

6.1. *Problem and parameterization*

EngiBench Beams2D v1 supplies a density-based finite-element model of the right half of a symmetric Messerschmitt-Bolkow-Blohm beam (Felten et al. 2025). Surrogate-assisted evolutionary optimization has previously been developed for expensive structural design (Yu, Li, and Liang 2020), and uncertainty-guided reliable-region search has been studied for expensive engineering optimization (Suprayitno and Yu 2019). Here, the beam provides engineering application validation of the audit framework within an expensive structural optimization workflow. The domain uses 100 by 50 bilinear elements. Horizontal degrees of freedom along the left symmetry boundary are fixed, the lower-right vertical degree of freedom is supported, and a unit downward load is applied at the top-left of the modeled half-beam. The material interpolation uses minimum stiffness 10ˆ-9, unit maximum stiffness, penalization power 3, and Poisson ratio 0.3. The objective is structural compliance; lower values are better.

Direct optimization of all 5,000 element densities is outside the studied surrogate's budget and initialization regime. A constrained engineering family is defined in which 19 effective logits plus one fixed anchor form a 4 by 5 coarse field. Cubic interpolation expands the field to 50 by 100 elements. A scalar logistic shift is solved by bisection so that every design has mean density 0.35. The study therefore evaluates model management within this smooth 19-dimensional design family rather than claiming general topology-optimization performance.

6.2. *Application protocol and audit results*

Before formal labels, five methods, ten repeats, and 200 evaluations per run are frozen. The first batch keeps the native 38-point initialization of GP-EI and TuRBO-1 while Greedy, fixed 01, and PEM-T share a 58-point initialization. A prespecified sensitivity batch then reruns GP-EI and TuRBO-1 from exactly the same 58 points and deterministic labels as the internal methods. Two acquisition controls, GP-PI and GP-UCB, and the author-code DRSO SAEA are subsequently run from the same archive. Their subsequent searches retain their stated acquisition, trust-region, relation-model, evolutionary, and candidate-generation rules. The original 50 trajectories contain 10,000 paid finite-element evaluations. The five shared-initialization external controls add 10,000 algorithm-budget

observations, reuse 2,900 initial labels, and make 7,100 additional finite-element calls. All 100 trajectories finish without failure or exact duplicate points. Independent endpoint diagnostics reproduce the original GP-EI and TuRBO-1 sensitivity endpoints within a maximum relative difference of 6.23e-12; event-level audits accept every GP-PI, GP-UCB, and DRSO trajectory.

Table 6. Finite-element endpoint compliance over ten optimization repeats.

Method	Median	Q1	Q3	Median improvement after shared initialization
Greedy	1445.1	1300.8	1694.9	0.362
Fixed01	1435.6	1354.2	1580.7	0.383
PEM-T	1972.8	1759.3	2168.9	0.251
GP-EI	2308.5	1989.4	2464.5	0.000
GP-PI	1233.1	921.2	1399.7	0.517
GP-UCB	2308.7	2223.8	2827.3	0.000
DRSO	647.4	580.4	780.0	0.747
TuRBO-1	574.0	534.7	665.3	0.751

PEM-T minus fixed 01 has median paired compliance difference +488.7, with interquartile range +134.7 to +814.7; PEM-T is better in two repeats and worse in eight. The corresponding comparison with greedy has median difference +575.2 and the same direction count. Under matched initialization, GP-EI minus PEM-T has median difference +242.8; GP-EI is better once, tied three times, and worse six times (paired Wilcoxon Holm-adjusted p=0.0469). Within the three added-control family, GP-PI minus PEM-T has median difference -876.1 and nine wins (Holm-adjusted p=0.0117), GP-UCB has median difference +574.6 with two wins, two ties, and six losses (p=0.0391), and DRSO has median difference -1334.6 with nine wins and one loss (p=0.0117). TuRBO-1 minus PEM-T has median difference -1372.3 and is better in all ten repeats (Holm-adjusted p=0.00391 in the original sensitivity family). Changing from native to shared initialization produces no supported endpoint shift for GP-EI (Holm-adjusted p=0.320) or TuRBO-1 (p=0.695). The application therefore distinguishes a weak acquisition setting from successful Bayesian, evolutionary-surrogate, and trust-region controls under the same starting evidence.

The shared-initialization finite-element trajectories and their between-repeat dispersion are shown in Figure 4.

Figure 4 separates the common initialization from method-specific search. All curves share the same values up to the dashed 58-evaluation boundary. TuRBO-1 and DRSO then improve rapidly and reach the lowest endpoint group. GP-PI also produces sustained improvement and finishes below the three internal methods, whereas GP-EI and GP-UCB show little median improvement after initialization. Greedy and fixed 01 improve more gradually and remain below PEM-T near the endpoint. The shaded interquartile bands preserve between-repeat variability and show that the endpoint ordering is not produced by a single trajectory.

Representative endpoint density fields for all eight shared-initialization methods are shown on the common material-density scale in Figure 5.

Figure 5 shows the median-endpoint design from every method on the common density scale. Because every field satisfies the same mean-density constraint, the contrast concerns material placement rather than material quantity. TuRBO-1 and DRSO form concentrated high-density load paths and reach the lowest compliance group. GP-PI

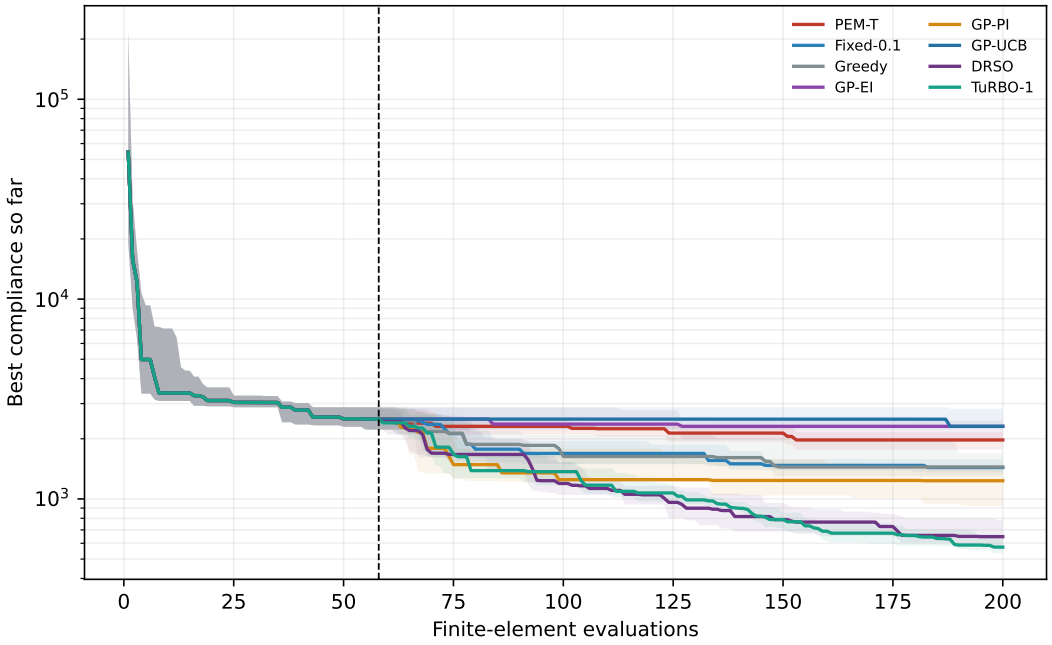


Figure 4. Median and interquartile best-so-far finite-element compliance under a shared 58-point initialization; the dashed line marks the end of initialization.

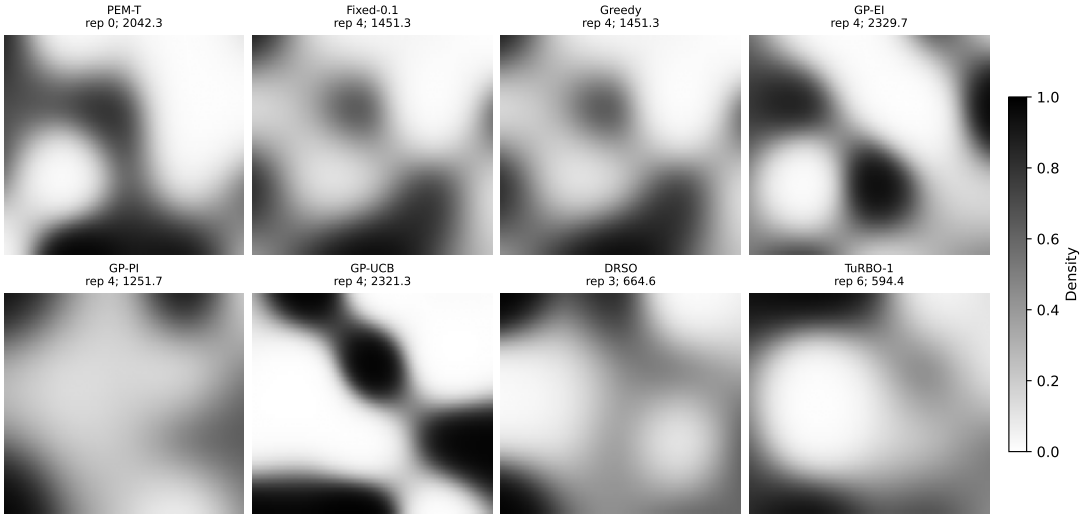


Figure 5. Representative median-run density fields for all eight methods under the shared-initialization comparison.

produces a stronger structural path than GP-EI or GP-UCB, whose diffuse fields remain close to their initial designs. The visual differences are consistent with the convergence pathways and endpoint ordering in Table 6.

6.3. *Conditional first-action records and budgeted replay*

The observed opportunity is $[m_0(5) - \min_j m_j(5)]/m_{\text{init}}$, including the baseline in the minimum. This describes the recorded branches, not a global optimum. At its original margin, archived mean continuation has above-margin opportunities in only two of thirty states and two of ten seed groups. Its 46 distinct nonbaseline contrasts have 3 better, 34 tie and 9 worse labels. In the independently acquired mean and EI records, best-observed improvement is exactly zero in every state, including below-margin gains. All original groups and margins remain retained; positive-case diagnosis cannot be validated by those independent outcomes.

The independent conditional record in Table 7 has 25 distinct nonbaseline-action contrasts per continuation: mean continuation gives 0 better, 24 tie and 1 worse; EI gives 0 better, 23 tie and 2 worse at its primary margin. These 30 source states share ten retained seed blocks, six of which contain nonbaseline-action contrasts. The immediate rule uses 42 requests and the completed-target rules 252 across those states; the immediate rule has 0/1 classification errors for mean/EI and zero reported state-selection loss for both. Completed-target zero errors follow from measuring the target. At budget 12, fixed depth completes 17/25 and adaptive 9/25 with 204 requests each; reversing alternative order preserves coverage but changes the evaluated class set in eight states and can leave one worse EI contrast unmeasured. Supplementary tables retain all existing budgets/margins, complete seed-block counts and the selected paired-grid ranges. The paired-grid ranges are preserved historical outputs; three final raw recovery files remain unavailable in the current preservation check. This limits complete independent raw-record replay for that supporting check, as detailed in the data-availability statement and supplement.

Table 7. Independent primary h5 record at margin 0.001. Truth B/T/W counts exclude exact baseline aliases and contain no better cases. Requests are charged across all 30 dependent states within ten seed blocks; they are not new physical solves. Completed-target errors are zero by direct measurement. Calibration is the constructed root-EI gate, not a published-method replication. This table was assembled with OpenAI Codex (GPT-6)-assisted Python and requires author verification.

Followup	Truth B/T/W	Rule	Classified	Errors	Requests	Selection loss (%)
MEAN	0/24/1	calibration	6/25	6	240	0.0000
MEAN	0/24/1	immediate	25/25	0	42	0.0000
MEAN	0/24/1	fixed depth	25/25	0	252	0.0000
MEAN	0/24/1	adaptive	25/25	0	252	0.0000
EI	0/23/2	calibration	6/25	6	240	0.2983
EI	0/23/2	immediate	25/25	1	42	0.0000
EI	0/23/2	fixed depth	25/25	0	252	0.0000
EI	0/23/2	adaptive	25/25	0	252	0.0000

The response-request totals above describe the audit reader. Executing the chosen branch in all 30 states additionally includes alias-only states: the original complete-target audit subtotal of 252 becomes 330, and the budget-12 subtotal of 204 becomes 282. These logical counts include previously acquired responses and do not measure additional physical solves. The subsequent monotone development schedules also require

explicit chosen-branch completion and a separate selection-certification count; their 13-cell reconciliation appears in the supplementary checked-records section.

6.4. *GP action-set certificates and computational comparison*

The separately frozen protocol selected two witnesses before either response and used its complete two-attempt allowance. Their true compliances were 14324.7435408532 and 9639.9247638608, versus stored two-vector lower bounds 232.5285149112 and 200.5953747176. Relative equilibrium residuals were below 1.1×10^{-10}. Both responses exceeded their root-tie thresholds. This identifies loose point bounds at those witnesses without disqualifying the full pool. No retry, replacement or optimization sample was acquired.

The GP construction retained all sixty metadata states and both complete ten-group datasets; seventeen alias-only states needed no bounds. The 43 nontrivial mean contexts have 114 first-action branches. Each next pool contains 1024 eligible actions. Both predeclared domains, $[0, \infty)$ and $[\lambda_{\text{pending}}, \infty)$, retain 2–7 actions, with medians four and three in the two datasets. Every stored next action is included. The original construction preceded reading first/next responses. The subsequent envelope comparison is explicitly post hoc development on its stored coefficient bands.

Table 8. Both GP response domains give the same recorded summaries. Complete metadata groups remain retained; H1 counts exclude baseline contrasts.

Dataset	Active states / branches	Retained min / median / max	Next-action inclusion	Joint lower / initial median	H1 category / selection increments
Archived mean	26 / 72	2 / 4 / 7	72/72	5.9786% to 11.3742%	0/46; 0/26
Previous holdout mean	17 / 42	2 / 3 / 6	42/42	9.2066% to 20.1462%	0/25; 0/17

All 204 frozen source hashes, 114 reconstructed binary kernel/cross systems and their coefficient enclosures match. The 232,564 original exclusions and an equal number of new exclusions pass exact rational replay. The envelope returns identical action sets for all 228 branch-domain cases, with exact feasible intervals contained in the previous outward intervals. Rechecking the tightened intervals leaves all 71 nonbaseline H1 contrasts per domain abstaining and no baseline selection certified.

The computational comparison times coefficient-to-set construction on the same machine and inputs, alternating algorithm order over three repetitions per case. Summed casewise medians are 19.2578 seconds for the frozen pairwise implementation and 6.0409 seconds for the exact envelope: 3.19-fold stage acceleration. Separate complete proof-replay passes cost 3.5787 and 2.6687 seconds, respectively; including those passes gives 2.62-fold acceleration. The replay passes are single measurements, and repetitions, domains and branches are dependent. These times exclude the common coefficient systems, paid baseline and physical-bound library. They do not establish industrial or end-to-end deployment speedup, nor comparison with an optimized library or knowledge-gradient policy.

The complete comparative development actually acquired 44,146 cached state/design bounds, plus the GP and proof work. Full-pool versus subset counts of116,850 and568 are logical requests; the subset figure is counterfactual because the complete library

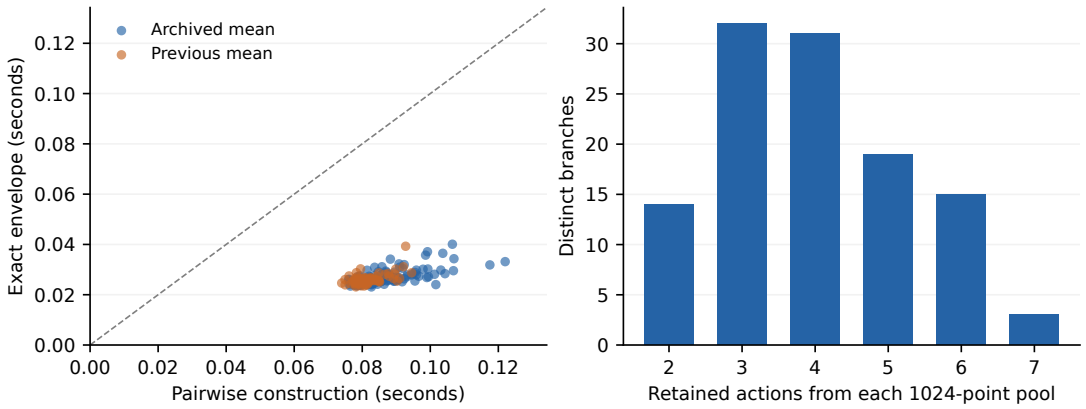


Figure 6. Paired median construction times and same-set cardinalities. The time panel uses all 228 dependent branch-domain cases; the cardinality panel counts each of114 branches once because both domains have the same counts. No independent optimization sample is represented.

was already acquired. Narrower action sets and faster certificate construction therefore establish a computational increment but do not repair the decision-sufficiency gate. H5 reachability and a useful H1 judgment remain unestablished. The original independent-optimization gate stays closed.

6.4.0.1. Which information blocks the engineering certificate?. Let $T = B - \epsilon$, where B is the paid H1 baseline and ϵ the declared absolute margin. For the existing relaxed lower-bound certificate in Eq. (3), $J \geq T$ holds if and only if $m_{\text{init}} \geq T$ and, for every retained action, both $\max(p, \inf R_i) \geq T$ and $\lambda_i \geq T$. This is a decomposition of the supplied certificate, not a new optimization theorem. A post hoc exact-rational check finds both information sources blocking all71 nonbaseline contrasts in each response domain. Of281 retained action terms per domain,71 pending-response terms and276 next-point bounds lie below their branch's threshold. The median, across contrasts, of the weakest retained next-point bound divided by its threshold is0.1622. Thus the present certificate cannot be repaired by increasing only one class of lower bounds while holding the relaxed action sets fixed. Changing the sets, exploiting coupled coefficient information or acquiring different physical information is outside this comparison. Failure of this sufficient certificate does not imply that the true action is inferior.

6.5. *Frozen-state intervention on GP acquisition policies*

To test the audit beyond the residual controller, thirty archived GP-EI, GP-PI and GP-UCB histories supply ten repeats per source policy. The states immediately before evaluations 59, 101, 151 and 200 are frozen, giving 120 interventions. Every fit uses only earlier labels. Logged ARD Matérn-5/2 parameters are replayed without refitting; all source-action means and standard deviations agree within the prespecified absolute and relative tolerance of 1e-8. Mean, EI, PI and LCB act on a shared 4,096-point Sobol pool augmented by the frozen source action. Here EI has zero improvement margin, PI uses 0.01 times training standard deviation, and LCB uses kappa=2. No intervention response is fed back into a source optimizer.

The 480 policy-state choices contain 261 distinct within-state actions. After choices have been frozen, 120 original action responses are legitimately reused; three further responses come from identical context-point pairs already evaluated in the diagnostic

ledger. The audit therefore acquires 138 new FE labels. A separate solver-compatibility check incurs 102 attempts, including an initial 42-call check whose numerical tolerance was found too strict. The final 60 retained comparisons pass the disclosed 1e-6 solver-relative tolerance, with maximum relative difference 4.929e-8. The 1e-8 GP replay tolerance is unchanged. Input hashes and choices were committed before new diagnostic acquisition; see S9 for the archive transfer and verification boundary.

Table 9. Acquisition-only intervention at frozen GP states. Results are means over ten repeat-level summaries, each averaging four checkpoints. U=max(0,training incumbent minus diagnostic compliance)/training standard deviation. Utility and paired differences are displayed in units of 1e-6; action-change percentages are relative to mean selection.

Frozen source	Control	Changed (%)	Mean U (1e-6)	Delta U vs mean (1e-6)
EI history	EI	30.0	0.899669	+0.000000
EI history	PI	25.0	0.899669	+0.000000
EI history	LCB	52.5	0.000000	-0.899669
PI history	EI	95.0	0.000000	-0.347088
PI history	PI	7.5	0.347088	+0.000000
PI history	LCB	97.5	0.000000	-0.347088
UCB history	EI	30.0	0.887232	+0.009475
UCB history	PI	35.0	1.178882	+0.301124
UCB history	LCB	75.0	0.511893	-0.365864

LCB often changes the selected point yet has negative mean immediate-utility differences in all three source-history groups. EI and PI effects depend on the state-generating history; a different choice does not automatically provide immediate gain. The small normalized effects must not be described as a large engineering improvement: training standard deviations range from 7.332e5 to 9.122e8, reflecting high-compliance designs. This is a bounded immediate diagnostic, not evidence for rejecting uncertainty-based exploration. The next experiment explicitly assesses longer horizons under different physical loads.

6.6. *Longer-horizon policies, load-position transfer and total cost*

The separately frozen protocol at commit 7555d58b145a2c44b73bab5ec47d0f643a57dde1 changes the top-edge downward-force position from 0.0 to 0.25 and 0.75. Each condition has ten fresh, paired Sobol initializations of 58 designs. Material, supports, 100-by-50 FE mesh, dimension 19 and volume fraction 0.35 are unchanged. These are two new physical conditions within one beam family, not independent engineering application domains.

All four transfer policies share an isotropic Matérn-5/2 GP and the same candidate seeds at a given phase and step. The GP is refitted with deterministic bounded L-BFGS-B, at most100 iterations and no restarts; alpha=1e-6, normalized targets, constant bounds[0.05,20], length-scale bounds[0.005,2], and initial length scale 0.2. A1,024-point Sobol pool excludes already observed points. EI and PI both use a 0.01-scaled improvement margin; LCB retains kappa=2. This controlled implementation differs from the frozen ARD posterior and larger augmented pool in Section 6.5, and from the historical GP optimizer. Consequently the new outcomes cannot be attributed solely to moving the force, or interpreted as exact reproductions of Table 6.

For each repeat the audited-selection workflow actually pays for 58 initial responses, four 20-step pilots, and 142 further evaluations of the pilot-selected policy. Its total

is 280 calls. Selection is written before validation; exact pilot-end ties use PI, mean, EI, LCB priority. The selected policy is validated from only the initial archive, with a candidate-seed namespace separate from the pilots. The comparison workflow separately pays for the same58 initial designs and all four 142-step continuations, totaling626 calls. Deterministic duplicates between workflows are queried again and counted. Each fixed member of the portfolio represents a 200-call alternative, not a free post hoc choice of its best member.

Every condition/policy has zero median improvement at its first pilot action, whereas20-step median improvement is positive. For LCB it is 2.514% and 3.649% in the two conditions; four of ten repeats in each condition improve after a zero-gain first action. This provides a concrete reason to keep delayed policy assessment separate from immediate screening. It measures diverging policy histories, not the isolated causal effect of a first action under a fixed continuation.

Table 10. Independent selected-policy validation and complete-portfolio comparison. Compliance is lower-is-better. Each compliance column is a median across ten repeats; the gap is the median of paired percent ratios, not a ratio of medians. Selected-policy validation uses 200 algorithm observations but 280 paid workflow calls including pilots; the portfolio pays 626.

Force position	Initial compliance	Selected-policy compliance	Portfolio-best compliance	Paired gap (%)	Exact best matches
0.25	1608.5	1172.5	943.2	9.014	4/10
0.75	700.7	406.4	320.6	27.217	3/10

The selected policy matches the independently executed portfolio-best endpoint in seven of twenty repeats. Its paired median gaps are 9.014% and 27.217%, and maxima83.067% and 289.223%. Pilot-end ties occur in seven and four repeats, so frequent PI selection is partly a consequence of the declared tie rule and must not be reported as strong empirical selection evidence. Compared with fixed PI, the selected workflow wins/ties/loses in 1/6/3 and 1/8/1 repeats; it makes 40% more calls than every fixed 200-call policy. Against the explicitly executed full portfolio it makes 55.272% fewer calls, accompanied by the disclosed endpoint losses. These counts do not establish universal savings, equivalence, or a performance-preserving deployment gate.

Figure 7 reports median initial-incumbent improvement and interquartile ranges together with the actual per-repeat call requirement. Selected-policy performance is the prespecified validation endpoint; pilot outcomes are excluded from its training. Portfolio performance is the best of all four paid members and is not a 200-call oracle baseline.

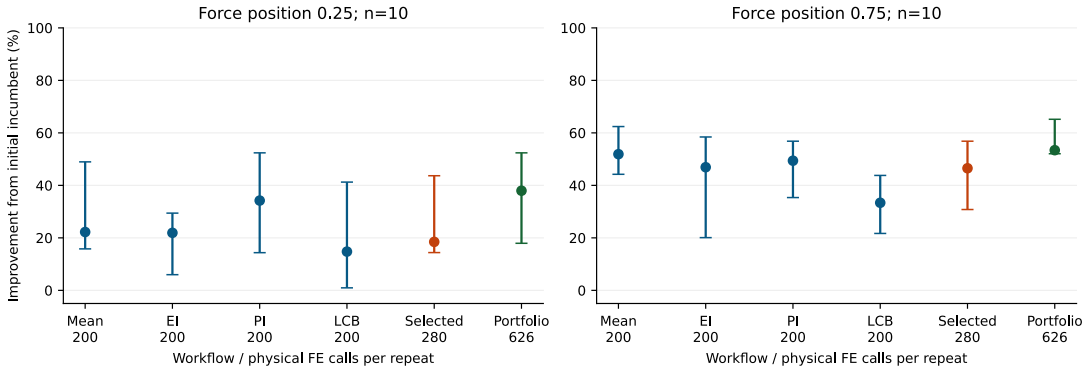


Figure 7. Cost and performance of the independently executed beam transfer workflows.

For practical design use, an engineer could retain a better design found in a pilot. A secondary analysis therefore also reports the best among every design paid for by the audited workflow. Its paired median portfolio gaps are 9.014% and 2.116%, respectively, compared with 9.014% and 27.217% for the prespecified selected-policy endpoint. Both endpoints are disclosed in S10; the secondary result does not replace the selected-policy test or create a retrospectively chosen promotion threshold. The difference shows why an audit must define the performance object as carefully as its cost boundary.

The retained transfer ledger contains 18,120 physical attempts and 20,000 full-trajectory algorithm observations across 100 budget 200 continuations, with shared initial observations counted inside each trajectory but paid only once per workflow. Loss of access to the previous complete local transfer journals required an explicit unchanged-protocol recovery; the earlier18,120 attempts are also counted. Together with the 138 immediate diagnostics and 102 solver checks, the earlier GP/transfer stages account for 36,480 new FE attempts. The reported optimizer and audit experiments account for 55,305 paid physics attempts. Subsequent auxiliary physical-bound development used 2 reference solves, bringing that earlier subtotal to 55,307. The separately frozen two-witness diagnosis adds two attempts, bringing the reported project total to 55,309; its GP construction and the new envelope verification add no physics. The opportunity, cost-reconciliation and assumption screens add no physical calls; their boundaries are reported in the supplement. Recovery responses enter each retained history once. Observed GP and FE component times from the retained execution are reported in S10 and are not exclusive resource measurements or demonstrated production wall-time savings. All new comparisons are descriptive and separate from the original prespecified inferential families.

Table 11. Additional released measured records. States, seed blocks, thermal cases and selected design pairs are separate reporting units. Paid solver attempts include recovery and verification; they are not pooled optimization-budget observations. The paired-grid record is fixed-design reevaluation, not fine-grid GP optimization. Tables were assembled using OpenAI Codex (GPT-6)-assisted Python code and remain subject to author verification.

Record	Unit	Measured result	Paid attempts
Archived common continuation	30 states within 10 shared seeds	90 named /46 distinct contrasts; 7 distinct changes in 4 states	282
Independent budget audit	10 new seed blocks; 30 dependent states	Budget 18: no allocation advantage; budget 12: depth 17/25, adaptive 9/25	2134
Small thermal verification	20 cases; 2 source positions	2 gate/simple disagreements: 1 tie and 1 loss	15888
Selected paired-grid check	14 contrasts; 9 existing states; 104 designs	Fixed h5 class stable in 14/14, including 8/8 differing endpoint pairs	521

7. Discussion and Design Guidance

The deterministic and empirical findings answer different questions. Ordering redundancy follows from the stated decision rule and holds whenever the correction is common across supported candidates. The observed performance loss is specific to the tested feedback controller, surrogate, candidates, and budgets. It does not imply that candidate-dependent uncertainty, conformal Bayesian optimization, or all adaptive exploration is ineffective.

The results support a staged design procedure for surrogate model management. First, identify the exact route from the monitoring statistic to the decision. If the signal adds a common constant before an argmin, it cannot alter ranking. Second, freeze

common states and compare the true utility of the alternative choices. This exposes an unhelpful exploration response before histories diverge. Third, run closed-loop validation on instances not used to select controls or coefficients. Calibration statistics remain useful diagnostics, but they should not serve as substitutes for either decision utility or final objective value.

The prototype illustrates why this sequence matters. Its failure fraction is close to 0.1, which could be read as successful monitoring. The same controller increases exploration from about 0.1 to 0.34-0.40, selects candidates with lower immediate improvement, and ends with larger errors. Localized weighting further improves some quantile-loss diagnostics without producing better trajectories. The missing link is not error measurement; it is a rule connecting error information to valuable evaluations.

For engineering users, the practical implication is to evaluate the acquisition pathway before paying for large simulation batches. A useful monitor should produce candidate-specific or management-level distinctions whose value can be tested, including delayed value for exploration, search-region changes, restarts, or stopping decisions. Monitoring width or error merely to increase exploration can consume the same number of expensive evaluations in less productive regions.

The added GP intervention tests the audit on actionable acquisition rules, while the two force-position conditions test finite-horizon policy and selection outcomes within the same beam family. The strongest reusable result is the evidence procedure: identify exact decision classes, pay only for undetermined immediate responses, preserve a label-free intervention boundary, and then disclose both closed-loop endpoints and total cost. The transfer selector illustrates a failure boundary rather than a new competitive optimizer. Its pilot ties, endpoint losses and extra cost relative to fixed policies delimit engineering deployment. The added thermal record uses a small authored heat-conduction model; industrial cross-domain validation and competitive warm-start selectors remain outside the evidence.

The study has limitations. Residuals originate from changing surrogate models and scales; no exchangeability argument establishes pointwise coverage. Sparse declarations limit conditional reliability statements. The RBF surrogate truncates training data, and the candidate process is one implementation among many. The broad BBOB external-reference batches retain their native initialization, although the finite-element sensitivity analysis removes this difference. The beam study uses a smooth 19-dimensional design family, not unrestricted 5,000-variable topology optimization. The independent budget record is offline sequential replay; its primary 18-request budget completes all distinct action classes, and source/calibration GP times were not separately recorded. Three-grid checks reevaluate recorded designs without refitting fine-grid GPs or refining every training point. Absolute compliance remains mesh/mapping-sensitive and the strict original coarse-response check flags one high-compliance point; neither mesh convergence nor physical accuracy of the illustrative 0.1% margin is certified. Full sensitivity, 300-dimensional, application, resource, and recovery evidence is supplied separately so that these limits remain auditable without dominating the main paper.

8. Conclusion

The conditional audit separates action identity, target-horizon consequences and the cost of obtaining a judgment. Its fixed-kernel GP component produces response-domain action sets with explicit coefficient enclosures and rational exclusion witnesses. The exact band envelope retains closed ties and constructs the relaxed-band solution in $O(M \log M)$

arithmetic operations. On the complete retained development cases, it preserves every original action set and stored next action, with 3.19-fold faster set construction than the frozen pairwise reference.

This computational result has a definite boundary. The original H1 point bounds remain too weak to produce a new category or baseline selection certificate, despite 2–7 retained actions from pools of1024. The primary independent H5 records contain no better alternative, and adaptive allocation does not improve on class-aware fixed depth. Complete baseline/alias execution and the acquired full-pool bound library must remain in the cost ledger. A smaller candidate set or a faster pruner is insufficient evidence of performance-preserving deployment savings.

The contributions are an executable certificate-producing operator, its scoped computational improvement and an engineering evaluation that exposes the remaining information bottleneck. Claims beyond the binary fixed-kernel model, native floating predictions for every response, physical margin accuracy, H5 reachable histories and independent positive-case efficacy require evidence outside this study. These limits make the present completed method evaluation falsifiable without changing historical protocols or adverse findings.

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Availability of Data and Materials

The source repository is private: https://github.com/ya-yu-zhang/Reliable-Surrogate-Assisted-Optimization-in-Expensive-Engineering-Design. A repository URL alone does not provide reviewer access. The separately prepared data-and-code packet contains the immutable GP execution archive, the exact selected baseline and source snapshots needed for its reconstruction, coefficient bands, original and envelope exclusion certificates, recorded timings, and verification code. These materials support zero-solver reproduction of the current action-set contribution. The 114 kernel systems and 228 branch-domain outputs have been checked against the archived bytes; retiming can change machine-dependent times and does not acquire independent optimization evidence.

The supporting paired-grid tables and historical release audits are preserved, but their complete raw reconstruction is currently unavailable. As of 14 October 2026, 22 of the 25 files listed in the paired-grid provenance audit have been recovered with matching SHA-256 hashes. The final recovery journal (`events.jsonl`), response collection (`responses.json`) and status (`run_status.json`) under `results/engineering_optimization_paired_mesh_recovery_v1/` have not been recovered. The earlier full review ZIP is not currently accessible. The retained analysis outputs cannot substitute for those raw records, and the historical release-validation result is not a claim of a complete current replay. Consequently, the paired-grid observations remain supporting historical results, with an explicit raw-evidence limitation; they do not establish mesh convergence or physical accuracy of the decision margin.

The authors report preserving the historical raw-output archive of approximately 13.6 GB locally. That archive has not been fully inspected in the present preparation and is not included in the scoped reviewer packet. Availability of all historical records is therefore not guaranteed by this statement. Reviewer materials must be supplied through the journal's accepted attachment or access route before evaluation. No new physics, witness retry, replacement, or independent optimization sample was used to prepare these materials. Third-party implementations and publisher templates retain their own terms; private source access does not by itself grant an open-source license.

Ethics Approval

Not applicable. This study did not involve human participants or animals.

Conflicts of Interest

The authors declare no conflicts of interest to report regarding the present study.

Supplementary Materials

The supplementary document contains the certified-envelope proof details and H1 cost boundaries as well as the historical function-level, LassoBench, dimensional, sensitivity and resource evidence, plus the new GP intervention, force-position transfer, per-repeat endpoints, fixed-policy comparisons, horizon diagnostics and complete recovery-aware cost ledger.

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